

GIS-Based School Site Selection in Lahore, Pakistan

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Abstract

Problem Statement

Lahore is one of the fastest growing cities in Pakistan, and the demand for private schools has been rising steadily over the years. More families are looking for better education options for their children, and public schools are simply not meeting that demand. This has created a real business opportunity for anyone thinking about opening a private school. But the problem is, most people who want to start a school do not really know where to set it up or how much money they will actually need. Picking the wrong area means fewer students, too much competition nearby, and possibly shutting down within a few years. This project tries to answer a very practical question: which areas in Lahore are actually good for opening a private school, and how much investment would be needed to get started there? These are the two things any serious investor needs to know before putting money into this kind of business.

Objectives

The main goals of this project are to find areas in Lahore where there are a lot of children but not enough schools nearby, to figure out how much it would cost to rent or buy property in those areas, to check whether roads and transport links make those areas easy to reach for students, to understand what kind of families live there so the school fee can be set at the right level, and finally to create a map that shows which locations make the most sense from a business point of view based on all these factors combined.

Proposed Methodology and Data Sources

The good thing about this project is that most of the data needed will be available. Population data will come from the Pakistan Bureau of Statistics 2023 census and WorldPop, which gives detailed population grids. School locations across Lahore will be taken from OpenStreetMap and the Punjab School Education Department. For road and travel-time analysis, OpenStreetMap road network data will be used through OSMnx. Property prices and rental rates will be collected from Zameen.com, which is Pakistan's biggest property website and has detailed area-level pricing for Lahore. City boundary and area shapefiles will be downloaded from LDA and GADM. The analysis will be done using QGIS and Python. The main spatial techniques include Kernel Density Estimation to see where students are concentrated, fixed-bandwidth kernel weights will be applied using a 1 to 2 kilometer distance threshold to construct a spatial weight matrix around existing school locations, identifying statistically underserved zones with minimal competitive density, and network analysis to measure travel time. All these factors will then be combined using Multi-Criteria Decision Analysis to give each area a score. On the financial side, setup costs including rent, renovation, furniture, and staff salaries will be estimated to calculate how many students a school would need just to break even.

Expected Outcomes

By the end of this project, there will be a clear map showing the best areas in Lahore to open a school, a cost comparison table for the top locations, and a break-even analysis telling an investor how many students they need to at least cover their costs. The idea is that someone who wants to open a school should be able to look at these results and make a confident, informed decision about where to invest and how much to budget.